Chapter 5. The Riemann–Stieltjes Integral

Riemann integral, integrability criteria, fundamental theorem, bounded variation, Stieltjes integrals

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5.1 The Riemann integral

Let $f : [a,b] \to \mathbb{R}$ be bounded. A **partition** $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$ has mesh $\|P\| = \max \Delta x_i$, $\Delta x_i = x_i - x_{i-1}$. With $M_i = \sup_{[x_{i-1},x_i]} f$ and $m_i = \inf_{[x_{i-1},x_i]} f$, the upper and lower sums are

$$U(f,P) = \sum M_i \Delta x_i, \qquad L(f,P) = \sum m_i \Delta x_i.$$

Proposition 5.1 (Refinement)

If $P \subset Q$, then $L(f,P) \le L(f,Q) \le U(f,Q) \le U(f,P)$. Hence $L(f,P) \le U(f,P')$ for any two partitions.

Proof. Adding one point splits one interval; the infimum over each half is at least the infimum over the whole, so L cannot decrease; similarly for U. For the second claim compare both with $P \cup P'$. ∎

Definition (Riemann integrability)

The lower and upper integrals are $\underline{\int} f = \sup_P L(f,P)$ and $\overline{\int} f = \inf_P U(f,P)$; always $\underline{\int} f \le \overline{\int} f$. f is **Riemann integrable**, $f \in \mathcal{R}[a,b]$, if they are equal; the common value is $\int_a^b f$.

Proposition 5.2 (Riemann's criterion)

$f \in \mathcal{R}[a,b]$ iff for every $\epsilon > 0$ there is a partition P with $U(f,P) - L(f,P) < \epsilon$.

Proof. If f is integrable, choose P_1, P_2 with $U(f,P_1) < \int f + \epsilon/2$, $L(f,P_2) > \int f - \epsilon/2$, and take $P = P_1 \cup P_2$. Conversely $0 \le \overline{\int} f - \underline{\int} f \le U(f,P) - L(f,P) < \epsilon$. ∎

The Dirichlet function (1 on $\mathbb{Q}$, 0 otherwise) is not integrable: every $U = 1$ and every $L = 0$.

Theorem 5.3 (Properties)

For $f, g \in \mathcal{R}[a,b]$ and $c \in \mathbb{R}$: $f+g$, $cf \in \mathcal{R}$ and the integral is linear; $f \le g$ implies $\int f \le \int g$; for $a < c < b$, f is integrable on $[a,c]$ and $[c,b]$ with $\int_a^b = \int_a^c + \int_c^b$; $|f| \in \mathcal{R}$ and $|\int f| \le \int |f|$; $fg \in \mathcal{R}$.

Proof. For sums, $\sup(f+g) \le \sup f + \sup g$ on each subinterval, so $U(f+g,P) \le U(f,P) + U(g,P)$, similarly for L, and Riemann's criterion applies. For $|f|$, $\sup|f| - \inf|f| \le \sup f - \inf f$ on each subinterval. For products, f^2 is integrable because $\sup f^2 - \inf f^2 \le 2K(\sup f - \inf f)$ with $K = \sup|f|$, and $fg = \frac{1}{4}\{(f+g)^2 - (f-g)^2\}$. Additivity: insert c into partitions. ∎

5.2 Integrable functions

Theorem 5.4

Continuous functions and monotone functions on $[a,b]$ are Riemann integrable.

Proof. Continuous: by uniform continuity (Theorem 3.9) choose δ with oscillation $< \epsilon/(b-a)$ on intervals shorter than δ; any P with $\|P\| < \delta$ gives $U - L < \epsilon$. Increasing: with the uniform partition into n pieces, $U - L = \frac{b-a}{n} \sum (f(x_i) - f(x_{i-1})) = \frac{b-a}{n}(f(b) - f(a)) \to 0$. ∎

More generally a bounded function with finitely many discontinuities is integrable (cover the discontinuities by short intervals), and if $f \in \mathcal{R}$ and ϕ is continuous on the range of f, then $\phi \circ f \in \mathcal{R}$.

(The complete answer — f is integrable iff its set of discontinuities has measure zero — is Lebesgue's criterion, Chapter 10.)

Riemann's original definition uses **Riemann sums** $S(f,P,\xi)=\sum f(\xi_i)\Delta x_i$ with tags $\xi_i\in[x_{i-1},x_i]$.

Theorem 5.5

For bounded f and $A\in\mathbb{R}$ the following are equivalent: (a) $f\in\mathcal{R}[a,b]$ and $\int f=A$; (b) for every $\epsilon>0$ there is $\delta>0$ such that $|S(f,P,\xi)-A|<\epsilon$ for all tagged partitions with $\|P\|<\delta$.

Proof. (b)⇒(a): choosing tags near the sup or inf on each piece shows $U(f,P)$ and $L(f,P)$ are within 2ϵ of A. (a)⇒(b): fix P_0 with $U(f,P_0)-L(f,P_0)<\epsilon$, having k points, and $|f|\le K$. If $\|P\|<\delta=\epsilon/(kK)$, at most k intervals of P contain points of P_0, contributing at most $2kK\delta=2\epsilon$ to $U(f,P)-L(f,P)$; the remaining intervals lie inside intervals of P_0 and contribute at most $U(f,P_0)-L(f,P_0)$. So $U(f,P)-L(f,P)<3\epsilon$, and every Riemann sum lies between $L(f,P)$ and $U(f,P)$. ∎

Example 5.1

$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{n^2+k^2}=\int_0^1\frac{dx}{1+x^2}=\frac{\pi}{4}$, recognizing a Riemann sum with mesh $1/n$.

5.3 The fundamental theorem of calculus

Theorem 5.6

For $f\in\mathcal{R}[a,b]$, $F(x)=\int_a^x f$ is Lipschitz, hence uniformly continuous. If f is continuous at x_0, then F is differentiable at x_0 with $F'(x_0)=f(x_0)$.

Proof. $|F(y)-F(x)|\le K|y-x|$ with $K=\sup|f|$. Next, $\frac{F(x_0+h)-F(x_0)}{h}-f(x_0)=\frac{1}{h}\int_{x_0}^{x_0+h}(f-f(x_0))$, which is at most $\sup_{|t-x_0|\le|h|}|f(t)-f(x_0)|\to 0$ in absolute value. ∎

So every continuous function has an antiderivative.

Theorem 5.7 (Fundamental theorem of calculus)

If F is differentiable on $[a,b]$ and $f=F'\in\mathcal{R}[a,b]$, then $\int_a^b f=F(b)-F(a)$.

Proof. For any partition, the mean value theorem gives $F(b)-F(a)=\sum F'(\xi_i)\Delta x_i$ for suitable ξ_i, a Riemann sum lying between $L(f,P)$ and $U(f,P)$. Hence $\underline{\int}f\le F(b)-F(a)\le\overline{\int}f$. ∎

The hypothesis $F'\in\mathcal{R}$ cannot be dropped: there are differentiable F with bounded but non-integrable derivatives (Volterra), and $x^2\sin(1/x^2)$ has an unbounded derivative.

Proposition 5.8 (Substitution and integration by parts)

(a) If $g:[a,b]\to[c,d]$ is C^1 and f is continuous on $[c,d]$, then $\int_a^b f(g(t))g'(t)\,dt=\int_{g(a)}^{g(b)}f(x)\,dx$.
(b) If F,G are differentiable with $F'=f$, $G'=g$ integrable, then $\int_a^b Fg=F(b)G(b)-F(a)G(a)-\int_a^b fG$.

Proof. (a) With $\Phi(y)=\int_{g(a)}^{y}f$, $(\Phi\circ g)'=f(g)g'$; apply Theorem 5.7. (b) Apply Theorem 5.7 to FG, whose derivative $fG+Fg$ is integrable. ∎

5.4 Functions of bounded variation

Definition (Bounded variation)

The **total variation** of $f:[a,b]\to\mathbb{R}$ is $V_a^b(f)=\sup_P\sum_i|f(x_i)-f(x_{i-1})|$. f is of **bounded variation** (BV) if $V_a^b(f)<\infty$.

Monotone functions are BV with $V=|f(b)-f(a)|$; Lipschitz functions are BV; $x\sin(1/x)$ is continuous on $[0,1]$ but not BV (the variation over the points $1/((k+\frac{1}{2})\pi)$ is a harmonic series). Sums and products of BV functions are BV, and $V_a^b=V_a^c+V_c^b$ for $a<c<b$ (refine partitions by c).

Theorem 5.9 (Jordan decomposition)

f is BV on $[a,b]$ iff $f=g-h$ with g,h increasing. One can take $g(x)=V_a^x(f)$ and $h=g-f$.

Proof. Differences of increasing functions are BV. Conversely $g(x)=V_a^x(f)$ increases, and for $x<y$, $h(y)-h(x)=V_x^y(f)-(f(y)-f(x))\geq 0$. ∎

Hence a BV function has one-sided limits everywhere and countably many discontinuities, and is Riemann integrable.

A curve $\gamma:[a,b]\to\mathbb{R}^m$ has length $\ell(\gamma)=\sup_P\sum\|\gamma(t_i)-\gamma(t_{i-1})\|$; it is **rectifiable** if this is finite. Since $|\gamma_j(t)-\gamma_j(s)|\leq\|\gamma(t)-\gamma(s)\|\leq\sum_j|\gamma_j(t)-\gamma_j(s)|$, γ is rectifiable iff every component is BV.

Theorem 5.10

A C^1 curve $\gamma:[a,b]\to\mathbb{R}^m$ is rectifiable with $\ell(\gamma)=\int_a^b\|\gamma'(t)\|dt$.

Proof. $\|\gamma(t_i)-\gamma(t_{i-1})\|=\|\int_{t_{i-1}}^{t_i}\gamma'\|\leq\int_{t_{i-1}}^{t_i}\|\gamma'\|$ gives $\ell\leq\int\|\gamma'\|$. Conversely, by uniform continuity of γ', for fine partitions γ' varies by less than ϵ on each piece, so $\|\gamma(t_i)-\gamma(t_{i-1})\|\geq(\|\gamma'(t_i)\|-\epsilon)\Delta t_i$, and summing gives $\ell\geq\int\|\gamma'\|-2\epsilon(b-a)$ in the limit. ∎

5.5 The Riemann–Stieltjes integral

Let $\alpha:[a,b]\to\mathbb{R}$ be increasing. Replace Δx_i by $\Delta\alpha_i=\alpha(x_i)-\alpha(x_{i-1})$: $U(f,\alpha,P)=\sum M_i\Delta\alpha_i$, $L(f,\alpha,P)=\sum m_i\Delta\alpha_i$. f is **Riemann–Stieltjes integrable** with respect to α, $f\in\mathcal{R}(\alpha)$, if $\inf_P U=\sup_P L$; the value is $\int_a^b f\,d\alpha$. Refinement, Riemann's criterion and linearity in f and in α carry over word for word; for BV $\alpha=\alpha_1-\alpha_2$ (increasing parts) one sets $\int f d\alpha=\int f d\alpha_1-\int f d\alpha_2$, which is equivalent to the Riemann-sum definition with mesh $\to 0$ when f is continuous.

Proposition 5.11

If f is continuous on $[a,b]$ and α is BV, then $f\in\mathcal{R}(\alpha)$, and $|\int f d\alpha|\leq\sup|f|\cdot V_a^b(\alpha)$.

Proof. For increasing α: choose δ from uniform continuity so that the oscillation of f on pieces shorter than δ is less than ϵ; then $U-L\leq\epsilon(\alpha(b)-\alpha(a))$. ∎

Theorem 5.12

(a) If α is C^1 and f is bounded, then $f\in\mathcal{R}(\alpha)$ iff $f\alpha'\in\mathcal{R}$, and $\int f d\alpha=\int f(x)\alpha'(x)dx$. (b) If α is a step function with jumps c_k at points s_k in the interior and f is continuous at each s_k, then $\int f d\alpha=\sum_k c_k f(s_k)$.

Proof. (a) By the mean value theorem $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$, so Riemann–Stieltjes sums of f and Riemann sums of $f\alpha'$ with the same tags differ by $\sum f(\xi_i)\{\alpha'(t_i) - \alpha'(\xi_i)\}\Delta x_i$, which is small by uniform continuity of α'. (b) Only the subintervals containing a jump contribute, and there $f \approx f(s_k)$. ∎

Thus sums and integrals are special cases of one object; in probability, $E\,g(X) = \int g\,dF$ for a distribution function F.

Theorem 5.13 (Integration by parts)

If f, α are BV and $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ and $\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a)$.

Proof. Summation by parts: for a tagged partition, $\sum \alpha(\xi_i)\{f(x_i) - f(x_{i-1})\} = f(b)\alpha(b) - f(a)\alpha(a) - \sum f(x_i)\{\alpha(\xi_{i+1}) - \alpha(\xi_i)\}$ (with $\xi_0 = a$, $\xi_{n+1} = b$), and the last sum is a Riemann–Stieltjes sum for $\int f d\alpha$ over the partition $\{\xi_i\}$, whose mesh is at most $2\|P\|$. ∎

Corollary 5.14 (Second mean value theorem)

If f is increasing and g is continuous on $[a, b]$, there is $c \in [a, b]$ with $\int_a^b f(x)g(x)dx = f(a)\int_a^c g + f(b)\int_c^b g$.

Proof. With $G(x) = \int_a^x g$, Theorem 5.12(a) and integration by parts give $\int fg = \int f dG = f(b)G(b) - \int_a^b G\,df$. Since $df \geq 0$ and G is continuous, $\int G df = G(c)(f(b) - f(a))$ for some c (intermediate value theorem). Rearranging gives the formula. ∎